NUMERICAL
MATRIX SOLVER

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Hardcoded Excel Direct Stiffness Method
with Array Partitioning Architecture

CONTENT & FINITE ELEMENT METHODOLOGY

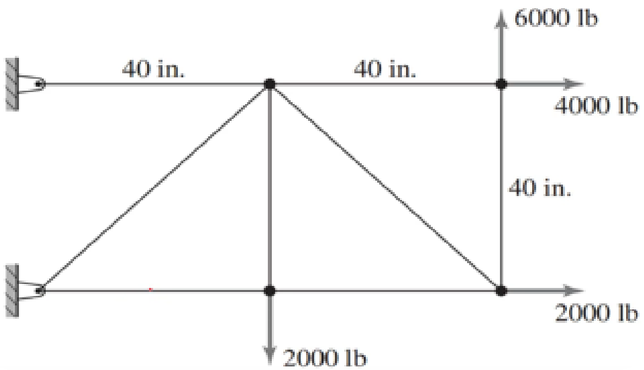
- Step 1: Discretize the physical truss structure into an organized grid of nodes and elements.
- Step 2: Formulate the local element stiffness matrices using material properties and orientation angles.
- Step 3: Assemble the local matrices into a centralized master twelve-by-twelve global stiffness matrix.
- Step 4: Partition the global matrix by separating the fixed boundary supports from the active free joints.
- Step 5: Solve the active partitioned system using matrix inversion to determine all unknown nodal displacements.
- Step 6: Compute support reaction forces, internal member strains, engineering stresses, and total axial loads.

Structural Domain Configuration

To evaluate the accuracy of the custom Finite Element Analysis (FEA) engine, a statically indeterminate 2D truss structure was adapted from David V. Hutton's textbook, Fundamentals of Finite Element Analysis.

The structural system is discretized into 6 distinct nodes and 8 individual truss elements, yielding a complete system of 12 global Degrees of Freedom (DoF).

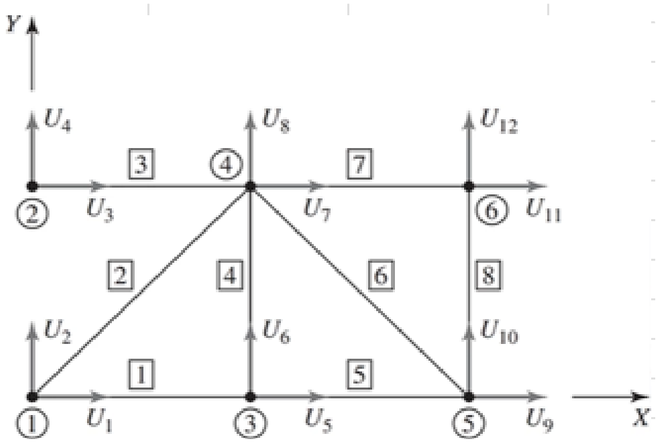
All structural elements are evaluated under identical material and cross-sectional parameters: a constant material Young's Modulus (E) of 1.00E+07 psi and a uniform cross-sectional area (A) of 1.50 in².



Member Geometric & Kinematic Discretization

Individual element geometry is established via local orientation angles relative to the global coordinate grid, determining directional cosines ($c = \cos(\theta)$) and sines ($s = \sin(\theta)$):

- Horizontal Members (Elements 1, 3, 5, 7):
  Orientation = 0.00 rad | L = 40.00 in | Stiffness k = 3.75E+05 lb/in.
  Axial stiffness operates entirely on the global X-axis ($c^2 = 1.00$).
- Vertical Members (Elements 4, 8):
  Orientation = 1.57 rad | L = 40.00 in | Stiffness k = 3.75E+05 lb/in.
  Stiffness maps completely to the global Y-axis ($s^2 = 1.00$).
- Diagonal Members (Elements 2, 6):
  Orientation = 0.79 rad / 5.50 rad | L = 56.57 in | Stiffness k = 2.65E+05 lb/in.
  Kinematics distribute evenly across both global axes ($c^2 = s^2 = 0.50$).



	θ(rad)	A	E	L	k	c^2	s^2	sc	$k*c^2$	$k*s^2$	k*sc
Element 1	0.00	1.50	1.00E+07	40.00	3.75E+05	1.00	0.00	0.00	3.75E+05	0.00E+00	0.00E+00
Element 2	0.79	1.50	1.00E+07	56.57	2.65E+05	0.50	0.50	0.50	1.33E+05	1.33E+05	1.33E+05
Element 3	0.00	1.50	1.00E+07	40.00	3.75E+05	1.00	0.00	0.00	3.75E+05	0.00E+00	0.00E+00
Element 4	1.57	1.50	1.00E+07	40.00	3.75E+05	0.00	1.00	0.00	1.41E-27	3.75E+05	2.30E-11
Element 5	0.00	1.50	1.00E+07	40.00	3.75E+05	1.00	0.00	0.00	3.75E+05	0.00E+00	0.00E+00
Element 6	5.50	1.50	1.00E+07	56.57	2.65E+05	0.50	0.50	-0.50	1.33E+05	1.33E+05	-1.33E+05
Element 7	0.00	1.50	1.00E+07	40.00	3.75E+05	1.00	0.00	0.00	3.75E+05	0.00E+00	0.00E+00
Element 8	1.57	1.50	1.00E+07	40.00	3.75E+05	0.00	1.00	0.00	1.41E-27	3.75E+05	2.30E-11

- The Design Inputs (Blue Columns - Modifiable Variables): The columns highlighted in blue (θ, A, E, L) act as the core user inputs. By altering the geometric cross-sectional areas (A), material constants (E), or node-to-node member lengths (L), the engineer can instantly evaluate different structural configurations or optimizations without rewriting a single equation.

- Automated Axial Stiffness Evaluation (k): The spreadsheet automatically runs a cell-by-cell calculation of the standard 1D truss axial stiffness: Stiffness Factor k = (A x E) / L This scales the raw material resistance against the physical dimensions of the member.

- Kinematic Local-to-Global Mapping Engine: The grey columns (c^2, s^2, sc, kc^2, ks^2, k*sc) represent the hardcoded trigonometric engine. It takes the orientation angle (Theta) of the element and converts its uniaxial stiffness vector into a 2D Cartesian plane.

$$[K^{(e)}] = k_e \begin{bmatrix} c^2 & sc & -c^2 & -sc \\ sc & s^2 & -sc & -s^2 \\ -c^2 & -sc & c^2 & sc \\ -sc & -s^2 & sc & s^2 \end{bmatrix}$$

Each of the eight blue-header blocks (K1 through K8) represents a single structural element. However, they are not just isolated parts anymore; they have been completely translated into the global coordinate system of the entire structure.

	1↓	2↓	5↓	6↓
	K1			
1→	3.75E+05	0.00E+00	-3.75E+05	0.00E+00
2→	0.00E+00	0.00E+00	0.00E+00	0.00E+00
5→	-3.75E+05	0.00E+00	3.75E+05	0.00E+00
6→	0.00E+00	0.00E+00	0.00E+00	0.00E+00

	1↓	2↓	7↓	8↓
	K2			
1→	1.33E+05	1.33E+05	-1.33E+05	-1.33E+05
2→	1.33E+05	1.33E+05	-1.33E+05	-1.33E+05
7→	-1.33E+05	-1.33E+05	1.33E+05	1.33E+05
8→	-1.33E+05	-1.33E+05	1.33E+05	1.33E+05

	3↓	4↓	7↓	8↓
	K3			
3→	3.75E+05	0.00E+00	-3.75E+05	0.00E+00
4→	0.00E+00	0.00E+00	0.00E+00	0.00E+00
7→	-3.75E+05	0.00E+00	3.75E+05	0.00E+00
8→	0.00E+00	0.00E+00	0.00E+00	0.00E+00

	5↓	6↓	7↓	8↓
	K4			
5→	1.41E-27	2.30E-11	-1.41E-27	-2.30E-11
6→	2.30E-11	3.75E+05	-2.30E-11	-3.75E+05
7→	-1.41E-27	-2.30E-11	1.41E-27	2.30E-11
8→	-2.30E-11	-3.75E+05	2.30E-11	3.75E+05

	5↓	6↓	9↓	10↓
	K5			
5→	3.75E+05	0.00E+00	-3.75E+05	0.00E+00
6→	0.00E+00	0.00E+00	0.00E+00	0.00E+00
9→	-3.75E+05	0.00E+00	3.75E+05	0.00E+00
10→	0.00E+00	0.00E+00	0.00E+00	0.00E+00

	9↓	10↓	7↓	8↓
	K6			
9→	1.33E+05	-1.33E+05	-1.33E+05	1.33E+05
10→	-1.33E+05	1.33E+05	1.33E+05	-1.33E+05
7→	-1.33E+05	1.33E+05	1.33E+05	-1.33E+05
8→	1.33E+05	-1.33E+05	-1.33E+05	1.33E+05

	7↓	8↓	11↓	12↓
	K7			
7→	3.75E+05	0.00E+00	-3.75E+05	0.00E+00
8→	0.00E+00	0.00E+00	0.00E+00	0.00E+00
11→	-3.75E+05	0.00E+00	3.75E+05	0.00E+00
12→	0.00E+00	0.00E+00	0.00E+00	0.00E+00

	9↓	10↓	11↓	12↓
	K8			
9→	1.41E-27	2.30E-11	-1.41E-27	-2.30E-11
10→	2.30E-11	3.75E+05	-2.30E-11	-3.75E+05
11→	-1.41E-27	-2.30E-11	1.41E-27	2.30E-11
12→	-2.30E-11	-3.75E+05	2.30E-11	3.75E+05

Notice the external numbers tracking along the top and left sides of each block (ranging from 1 to 12). These are the global addresses (Degrees of Freedom). They act like a precise routing index, showing exactly which structural joints move and where that specific element's stiffness pushes back against the system.

- The 12x12 Global Framework: This massive grid is the master global stiffness matrix. By reading the address tags from all eight individual elements, the spreadsheet automatically layers them on top of each other. Where multiple members meet at a single shared joint, their stiffness values sum together perfectly to represent the unified strength of the truss.

- The Partitioning Lines: The solid red cross lines running through the matrix are not just decorative—they represent your physical boundary conditions. Because the truss is pinned solidly to the wall at Nodes 1 and 2, global movements 1, 2, 3, and 4 are locked at absolute zero. The red lines divide the system mathematically to isolate what is fixed from what is free to move.

- Kcc (Top-Left): Represents the isolated stiffness links between the fixed support points themselves.
- Kaa (Bottom-Right): This 8x8 block is the most critical zone. It captures the pure structural stiffness of the completely free nodes (Movements 5 through 12) under loading.
- Kca and Kac (Off-Diagonals): These zones handle the geometric cross-talk, tracking how a movement at a free joint transfers stress back into the fixed wall supports.

Kcc Kca

	K											
	1	2	3	4	5	6	7	8	9	10	11	12
1	5.08E+05	1.33E+05	0.00E+00	0.00E+00	-3.75E+05	0.00E+00	-1.33E+05	-1.33E+05	0.00E+00	0.00E+00	0.00E+00	0.00E+00
2	1.33E+05	1.33E+05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	-1.33E+05	-1.33E+05	0.00E+00	0.00E+00	0.00E+00	0.00E+00
3	0.00E+00	0.00E+00	3.75E+05	0.00E+00	0.00E+00	0.00E+00	-3.75E+05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
4	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
5	-3.75E+05	0.00E+00	0.00E+00	0.00E+00	7.50E+05	2.30E-11	-1.41E-27	-2.30E-11	-3.75E+05	0.00E+00	0.00E+00	0.00E+00
6	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.30E-11	3.75E+05	-2.30E-11	-3.75E+05	0.00E+00	0.00E+00	0.00E+00	0.00E+00
7	-1.33E+05	-1.33E+05	-3.75E+05	0.00E+00	-1.41E-27	-2.30E-11	1.02E+06	2.30E-11	-1.33E+05	1.33E+05	-3.75E+05	0.00E+00
8	-1.33E+05	-1.33E+05	0.00E+00	0.00E+00	-2.30E-11	-3.75E+05	2.30E-11	6.40E+05	1.33E+05	-1.33E+05	0.00E+00	0.00E+00
9	0.00E+00	0.00E+00	0.00E+00	0.00E+00	-3.75E+05	0.00E+00	-1.33E+05	1.33E+05	5.08E+05	-1.33E+05	-1.41E-27	-2.30E-11
10	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.33E+05	-1.33E+05	-1.33E+05	5.08E+05	-2.30E-11	-3.75E+05
11	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	-3.75E+05	0.00E+00	-1.41E-27	-2.30E-11	3.75E+05	2.30E-11
12	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	-2.30E-11	-3.75E+05	2.30E-11	3.75E+05

Kac Kaa

	U			F	
Uc	0	U1	Fc	F1	
	0	U2		F2	
	0	U3		F3	
	0	U4		F4	
X	U5		=	0	F1
	U6			-2000	F2
	U7			0	F3
	U8			0	F4
Ua	U9		Fa	2000	F5
	U10			0	F6
	U11			4000	F7
	U12			6000	F8

The Interactive Loading Engine (The Blue Inputs)

- Modifiable Force Vector (Fa): The blue-highlighted column on the far right represents your active external loading vector. This column is fully modifiable and acts as your design control panel:
  - Entering -2000 at line 6 applies the vertical downward load on Node 3.
  - Entering 2000, 4000, and 6000 on lines 9, 11, and 12 instantly sets up the rest of the workspace.
- Live Automated Solving: Because the matrix is fully linked via active formulas, an engineer can change any value in this blue loading column to instantly re-simulate different real-world weather or weight conditions. The spreadsheet will instantly re-calculate the entire structural response without needing a single formula to be re-written.

$$\{U_a\} = [K_{aa}]^{-1}(\{F_a\} - [K_{ac}]\{U_c\})$$

	Ua
U5	0.0213333
U6	0.0408366
U7	-0.0160000
U8	0.0461699
U9	0.0426667
U10	0.1500914
U11	-0.0053333
U12	0.1660914

=

[Kaa]^-1							
2.67E-06	1.19E-21	-6.11E-22	1.27E-21	2.67E-06	2.67E-06	-6.11E-22	2.67E-06
8.83E-22	1.29E-05	-2.67E-06	1.02E-05	1.93E-21	1.29E-05	-2.67E-06	1.29E-05
-4.12E-22	-2.67E-06	2.67E-06	-2.67E-06	-8.23E-22	-5.33E-06	2.67E-06	-5.33E-06
1.05E-21	1.02E-05	-2.67E-06	1.02E-05	2.09E-21	1.29E-05	-2.67E-06	1.29E-05
2.67E-06	2.54E-21	-1.22E-21	2.54E-21	5.33E-06	5.33E-06	-1.22E-21	5.33E-06
2.67E-06	1.29E-05	-5.33E-06	1.29E-05	5.33E-06	3.11E-05	-5.33E-06	3.11E-05
-3.74E-22	-2.67E-06	2.67E-06	-2.67E-06	-7.48E-22	-5.33E-06	5.33E-06	-5.33E-06
2.67E-06	1.29E-05	-5.33E-06	1.29E-05	5.33E-06	3.11E-05	-5.33E-06	3.38E-05

X

Fa
0
-2000
0
0
2000
0
4000
6000

-

Kac			
-3.75E+05	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
-1.33E+05	-1.33E+05	-3.75E+05	0.00E+00
-1.33E+05	-1.33E+05	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00

X

Uc
0
0
0
0

$$\{F_c\} = [K_{cc}]\{U_c\} + [K_{ca}]\{U_a\}$$

	Fc
F1	-12000.00
F2	-4000.00
F3	6000.00
F4	0.00

=

Kcc			
5.08E+05	1.33E+05	0.00E+00	0.00E+00
1.33E+05	1.33E+05	0.00E+00	0.00E+00
0.00E+00	0.00E+00	3.75E+05	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00

X

Uc
0
0
0
0

+

Kca							
-3.75E+05	0.00E+00	-1.33E+05	-1.33E+05	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	-1.33E+05	-1.33E+05	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	-3.75E+05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

X

Ua
0.0213333
0.0408366
-0.0160000
0.0461699
0.0426667
0.1500914
-0.0053333
0.1660914

- The Active Partition Formula: The upper section executes the partitioned matrix equation to isolate and calculate unknown active translations. Because the support constraints are fixed at zero, the expression effectively reduces to multiplying the inverted active stiffness matrix by the applied external force vector.

- Automated Matrix Inversion: The large 8x8 block represents the inverted active stiffness matrix computed using Excel's native array functionalities. This operation yields the precise engineering displacements under the applied loading conditions. For example, Node 3 displaces horizontally by 0.021333 inches (U5) and vertically by 0.040836 inches (U6).

- Support Force Evaluation: Once all nodal translations are fully determined, the lower matrix line calculates the unknown reactions developing at the pinned boundaries (F1 through F4). The engine multiplies the connection zone matrix against the newly found displacement column.

- Numerical Force Results: The calculations yield the exact counter-balancing forces holding the structure in place against the external work loads:
- F1 (Node 1 Horizontal Reaction): -12,000.00 lb
- F2 (Node 1 Vertical Reaction): -4,000.00 lb
- F3 (Node 2 Horizontal Reaction): 6,000.00 lb
- F4 (Node 2 Vertical Reaction): 0.00 lb

EQUILIBRIUM VERIFICATION		
Sum of all forces in the global X direction	0.00	OK
Sum of all forces in the global Y direction	0.00	OK

- Static Equilibrium Check: To guarantee the absolute correctness of the mathematical solver, an automated validation script runs at the base of the sheet. It sums all active external forces and newly extracted support reactions across the global Cartesian axes.
- The Final Verification:
- Sum of all forces in the global X direction: 0.00 (Returns a green OK flag)
- Sum of all forces in the global Y direction: 0.00 (Returns a green OK flag)
- This absolute net-zero cancellation proves that the structural model perfectly satisfies the laws of static equilibrium.

	u1	u2	u2-u1	L	Strain	E	Stress	Force
Element 1	0.00	0.02	0.02133	40.00	5.33E-04	1.00E+07	5333.33	8000.00
Element 2	0.00	0.02	0.02133	56.57	3.77E-04	1.00E+07	3771.24	5656.85
Element 3	0.00	-0.02	-0.01600	40.00	-4.00E-04	1.00E+07	-4000.00	-6000.00
Element 4	0.04	0.05	0.00533	40.00	1.33E-04	1.00E+07	1333.33	2000.00
Element 5	0.02	0.04	0.02133	40.00	5.33E-04	1.00E+07	5333.33	8000.00
Element 6	-0.08	-0.04	0.03200	56.57	5.66E-04	1.00E+07	5656.85	8485.28
Element 7	-0.02	-0.01	0.01067	40.00	2.67E-04	1.00E+07	2666.67	4000.00
Element 8	0.15	0.17	0.01600	40.00	4.00E-04	1.00E+07	4000.00	6000.00

$$\varepsilon^{(e)} = \frac{du^{(e)}(x)}{dx} = \frac{d^{(e)}}{dx}[N_1(x) \quad N_2(x)] \begin{Bmatrix} u_1^{(e)} \\ u_2^{(e)} \end{Bmatrix}$$

$$= \begin{bmatrix} \frac{-1}{L^{(e)}} & \frac{1}{L^{(e)}} \end{bmatrix} \begin{Bmatrix} u_1^{(e)} \\ u_2^{(e)} \end{Bmatrix} = \frac{u_2^{(e)} - u_1^{(e)}}{L^{(e)}}$$

$$\sigma^{(e)} = E\varepsilon^{(e)}$$

Local Displacement Extraction

- Isolating Member Deformations: The table begins by extracting the localized start and end node displacements for each individual member, tracking how much the ends of each bar move relative to each other along their own axial path.
- Net Relative Displacement: By subtracting the start node displacement from the end node displacement, the spreadsheet calculates the true net elongation or shortening of the member. A positive change indicates the element is being pulled apart, while a negative value shows it is being compressed.

Mechanical Strain and Stress Calculations

- Automated Strain Tracking: The spreadsheet calculates the mechanical strain by taking the net relative displacement and dividing it by the total physical length of the member. This provides a dimensionless ratio representing the deformation intensity within the material.
- Material Stress Evaluation: Following the laws of material behavior, the engineering stress is found by multiplying the calculated strain directly by the material elasticity constant (E).
  - Tension Members: Elements like Member 1 and Member 2 show positive stress values (such as 5333.33 psi and 3771.24 psi), confirming they are operating in pure tension.
  - Compression Members: Member 3 shows a negative stress value of -4000.00 psi, proving that it acts as a structural strut carrying clamping or squeezing loads.

Internal Axial Forces

- Total Member Load Output: The final column calculates the true internal axial force traveling through each rod by multiplying the material stress by the cross-sectional area of the bar.
- Live Design Integration: Because this entire row-by-row inventory is dynamically tied to the very first parameter spreadsheet, any changes to the structural geometry or loading conditions will instantly ripple through to these final stress numbers. This allows the designer to verify immediately whether any member exceeds the safe allowable strength of the material.

aqus/CAE Learning Edition 2025 - Model Database: D:\Abaqus_Projects\04_2D_Truss\2D2D.cae [Viewport: 1]

ile Model Viewport View Result Plot Animate Report Options Tools Plug-ins Help

RT RT2

Module: Visualization Model: D:/Abaqus_Projects/04_2D_Truss/2d_truss.odb

Visualization defaults

Query

General Queries

- Node
- Distance
- Angle
- Mesh
- Element
- Mass properties
- Element face normal
- Advanced distance

Visualization Module Queries

- Probe values
- Stress linearization
- Active elements
- Active nodes
- Ply stack plot

Probe Values

Field Output...

Step: 1, Load Frame: 1

Field output variable for Probe: **RT, RT2**

Probe Values

Select from viewport | Key-in label | Select a display group

Probe: Nodes Components: Selected

Value for Attached elements:

Part Instance	Node ID	Orig. Coords	Def. Coords	Attached elements	RT, RT2
TRUSS_2D-1	5	0, 0, 0	1.22564e-32, 4	5, 7	-4000
TRUSS_2D-1	6	0, 39, 0	-6.25641e-33,	8	-0

Note: Click on respective check button to annotate values in viewer

Write to File... Cancel

2025 Tue Jul 21 12:33:05 Turkey Standard Time 20

```
The job input file "2d_truss.inp" has been submitted for analysis.
Job 2d_truss: Analysis Input File Processor completed successfully.
Job 2d_truss: Abaqus/Standard completed successfully.
Job 2d_truss completed successfully.
The job input file "2d_truss.inp" has been submitted for analysis.
Job 2d_truss: Analysis Input File Processor completed successfully.
Job 2d_truss: Abaqus/Standard completed successfully.
Job 2d_truss completed successfully.
```